



Data Professional - Mini Projects

Beginner Level: Power BI & SQL Fundamentals (Ideas 1 to 6)

1. Sales Data Exploration with SQL

- o *Concepts used:* SELECT, WHERE, ORDER BY, DISTINCT, TOP / LIMIT, aliases.
- o *Idea:* Query a sales table to answer 10 business questions (top 10 customers, orders by date, best-selling products).

2. Employee Summary Report

- o *Concepts used:* Aggregate functions (COUNT, SUM, AVG, MAX / MIN), GROUP BY, HAVING.
- o *Idea:* Summarize employee data by department: headcount, average salary, highest paid, and departments above a threshold.

3. Online Store Database with Joins

- o *Concepts used:* Primary / foreign keys, INNER / LEFT JOIN, basic normalization, CREATE TABLE & INSERT.
- o *Idea:* Design 4 related tables (customers, products, orders, order items) and write join queries for a full order report.

4. First Power BI Report: Sales Overview

- o *Concepts used:* Get Data (Excel / CSV), data types, cards, bar / line charts, slicers, formatting.
- o *Idea:* Import a sales file and build a one-page report with KPI cards, charts and slicers.

5. Data Cleaning with Power Query

- o *Concepts used:* Power Query Editor: remove duplicates, fix data types, split / merge columns, replace values, handle nulls.
- o *Idea:* Clean a messy customer file step by step, then load it into Power BI and compare before / after.

6. Sales Dashboard Connected to SQL

- o *Concepts used:* Power BI connection to SQL Server (Import mode), relationships, basic DAX measures (SUM, COUNT, DIVIDE).
- o *Idea:* Connect to the store database from idea 3 and build an interactive dashboard with KPI cards, 3 visuals and slicers.